(i) $(\angle 1+\angle 2)+(\angle 3+\angle 4)=180^\circ$ (Sum of opposite angles B and D)

(ii) $(\angle 1+\angle 4)+(\angle 2+\angle 3)=180^\circ$ (Sum of opposite angles A and C)

Now the result is given as follows.

Theorem 12 Opposite angles of a cyclic quadrilateral are supplementary.

Let us see the converse of theorem 12, which is very useful in solving problems

Converse of Theorem 12 If a pair of opposite angles of a quadrilateral is supplementary, then the quadrilateral is cyclic.

Activity - 7

Procedure

1. Draw a circle of any radius with centre O.
2. Mark any four points A, B, C and D on the boundary. Make a cyclic quadrilateral $ABCD$ and name the angles as in Fig. 4.78
3. Make a replica of the cyclic quadrilateral $ABCD$ with the help of tracing paper.
4. Make the cutout of the angles A, B, C and D as in Fig. 4.79
5. Paste the angle cutout $\angle 1, \angle 2, \angle 3$ and $\angle 4$ adjacent to the angles opposite to A, B, C and D as in Fig. 4.80
6. Measure the angles $\angle 1+\angle 3,$ and $\angle 2+\angle 4$.

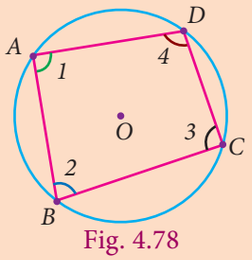


Fig. 4.78

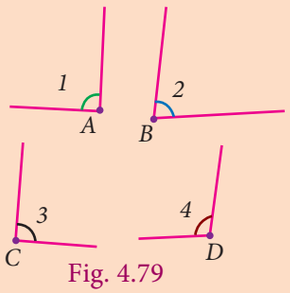


Fig. 4.79

Observe and complete the following:

1. (i) $\angle A+\angle C=$ _____ (ii) $\angle B+\angle D=$ _____

   (iii) $\angle C+\angle A=$ _____ (iv) $\angle D+\angle B=$ _____
2. Sum of opposite angles of a cyclic quadrilateral is _____.
3. The opposite angles of a cyclic quadrilateral is _____.

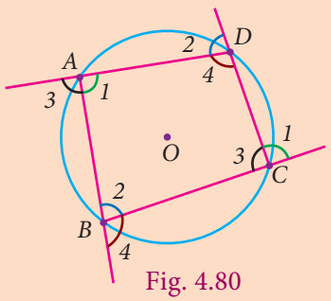


Fig. 4.80

Example 4.10 If $PQRS$ is a cyclic quadrilateral in which $\angle PSR=70^\circ$ and $\angle QPR=40^\circ$, then find $\angle PRQ$ (see Fig. 4.81).

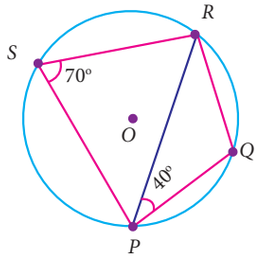


Fig. 4.81

Solution

$PQRS$ is a cyclic quadrilateral

Given $\angle PSR=70^\circ$

4-Geometry_Term1.indd 171 02/12/2022 21:42:43